

# Cortical Microcircuit Connectivity Generator

## Technical Documentation

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### Abstract

This document outlines the technical specification and implementation details for the generation of binary adjacency matrices representing synaptic connections within 3D cortical microcircuits. The system utilizes biologically realistic, distance-dependent probability rules optimized for high-performance computing.

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# 1 Overview

The primary objective of this module is to simulate the structural connectivity of a cortical microcircuit. Rather than assuming uniform connectivity, the generator applies specific spatial decay rules (Gaussian or Exponential) based on pre-synaptic and post-synaptic morphological types (m-types).

## 2 Function Reference

### 2.1 Function Signature

```
1 def build_adjacency_matrix(cell_mtypes, cell_coords, conn_data):
2     """
3     Generates a binary adjacency matrix representing synaptic
4     connections within a 3D cortical microcircuit.
5     """
```

Primary Interface

### 2.2 Arguments

- **cell\_mtypes (numpy.ndarray)**: A 1D array of length  $N$  containing the string morphological type (m-type) for each simulated cell (e.g., 'L23\_PC', 'L4\_SS').
- **cell\_coords (numpy.ndarray)**: An  $N \times 3$  array containing the spatial coordinates  $(x, y, z)$  for every cell in the microcircuit.
- **conn\_data (dict)**: A nested dictionary containing the empirical connectivity blueprints. This includes spatial fit parameters (a0mat\_exp, lmat\_gauss, etc.) and best-fit indicators for each pre/post m-type pathway.

### 2.3 Returns

- **adj\_matrix (numpy.ndarray)**: An  $N \times N$  binary matrix where 1 indicates a directed synaptic connection from the row index (pre-synaptic) to the column index (post-synaptic), and 0 indicates no connection.

## 3 Implementation Details

The generation process is highly optimized using vectorized operations to handle tens of thousands of neurons efficiently.

1. **Vectorized Distance Calculation:** The system computes the full  $N \times N$  3D Euclidean distance matrix instantly using `scipy.spatial.distance.cdist`, bypassing slow nested loops.
2. **Pathway-Specific Rules:** Cells are grouped by unique m-types. The algorithm iterates through pre/post pathways and applies the designated spatial decay formula:
  - *Exponential Decay:*  $P(d) = a_0 \cdot \exp(-d/\lambda)$
  - *Gaussian Decay:*  $P(d) = a_0 \cdot \exp(-d^2/(2\sigma^2))$

3. **Autapse Prevention:** The diagonal of the probability matrix is automatically zeroed out to ensure neurons do not form self-connections.
4. **Stochastic Generation:** The calculated probability matrix is compared against a matrix of uniformly distributed random numbers to determine the final binary state of each connection.

## 4 Getting Started

### 4.1 Prerequisites

Ensure the following dependencies are installed in your Python environment:

- `numpy`
- `scipy`

### 4.2 Installation

```
git clone https://github.com/username/cortical-connectivity.git
cd cortical-connectivity
pip install -r requirements.txt
```

## 5 License

This project is licensed under the MIT License - see the LICENSE file for details.

## 6 Contact

For technical inquiries or contributions, please contact the project maintainers via the GitHub repository or at [support@example.com](mailto:support@example.com).